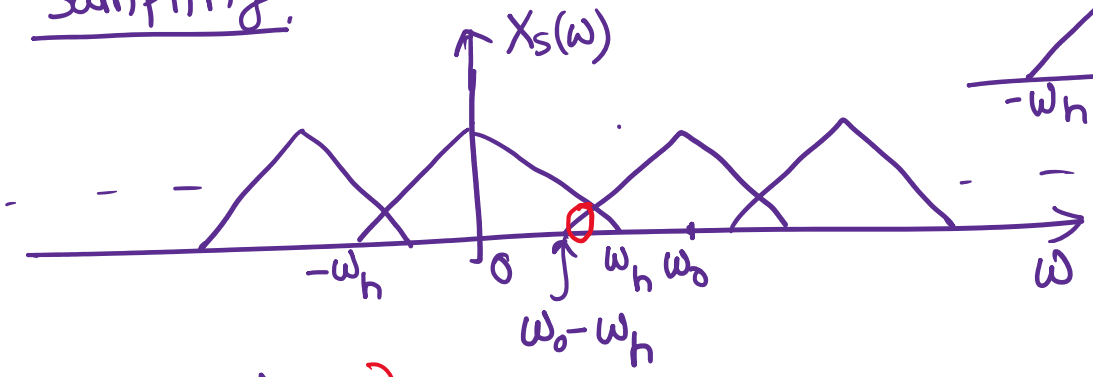
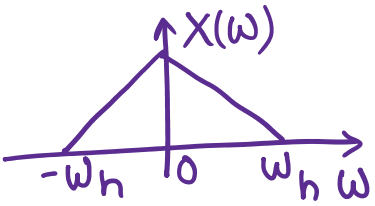
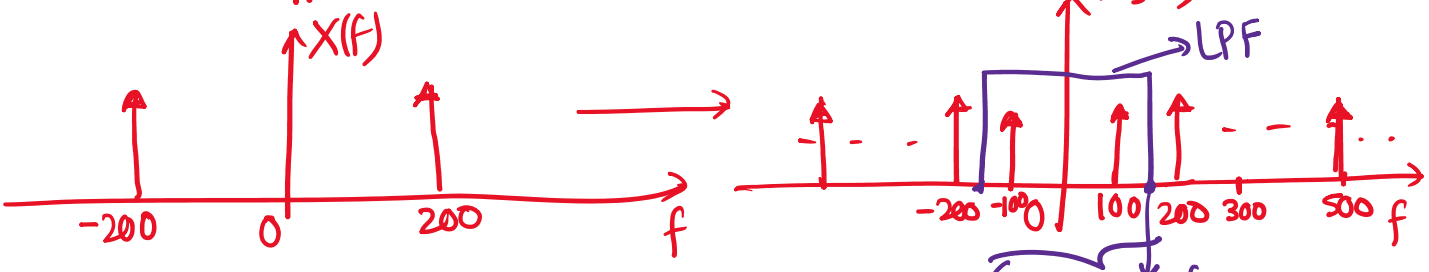
Sampling:





i.e. $\omega_0-\omega_h \leq \omega_h$
$\Rightarrow \omega_0 \leq 2\omega_h$ } Aliasing Distortⁿ

Ex.1) Let $\underbrace{x(t)=\cos(400\pi t)}_{f_h = 200\,Hz}$ & $f_s = 300\,Hz$



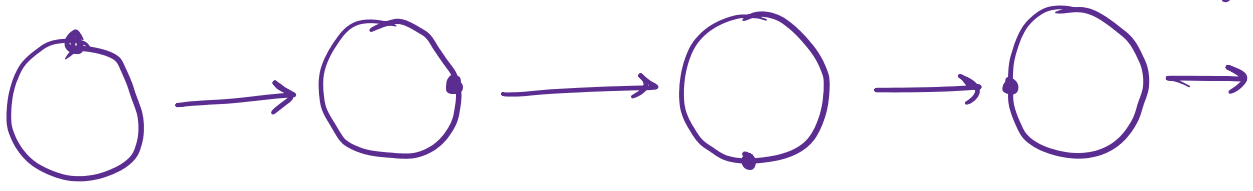
$f_c = \frac{f_s}{2} = 150\,Hz$

$x_r(t) = \cos(200\pi t) \rightarrow 100\,Hz$

$\therefore$ 200 Hz $\xrightarrow{\text{Alias}}$ 100 Hz
(Due to undersampling)

Ex.2) Sampling rate: 60 fps
Wheel rotⁿ (clkwise): (a) 30π rad/s, (b) 90π rad/s. (c) Stationary?

(a)



(Clockwise)

(b) $\Rightarrow$ Anticlkwise

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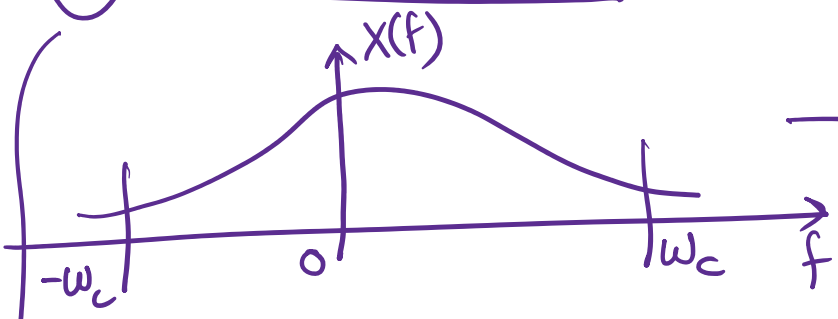
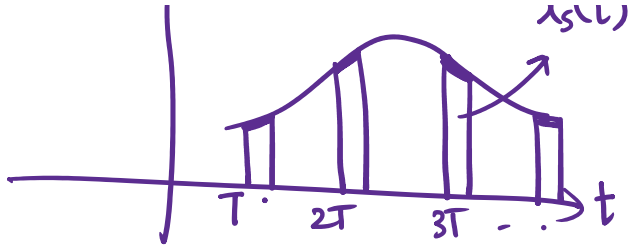
(2) Natural Sampling:

$x(f)$

(2) Natural sampling.



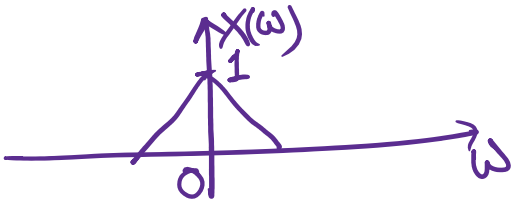


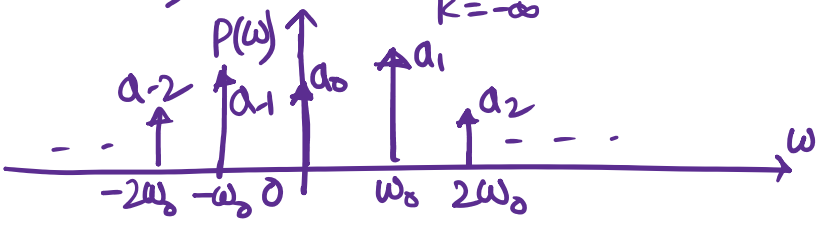
$$x_s(t) = x(t)p(t)$$

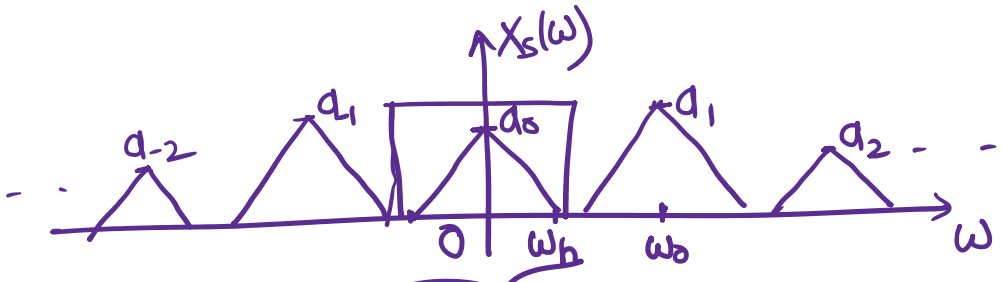
$$p(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$\Rightarrow X_s(\omega) = \frac{1}{2\pi} X(\omega) * P(\omega)$$

$$\Rightarrow P(\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_0)$$







LPF $\Rightarrow x(t)$, provided $\omega_0 > 2\omega_h$

Any $p(t)$ (periodic with T) works, provided $a_0 \neq 0$